Selected answers: Calculus Examination June 2016

Section A Multichoice

1 [c] 2 [e] 3 [b] 4 [d]
5 [d] 6 [c] 7 [b] 8 [b]
9 [d] 10 [c]

Section B

1.1 $(2,3)$ 1.2 $k=4$ 1.3 15

1.4 ∞ 1.5 centre $(1,2,-4)$, $r=6$

1.6 $\frac{1}{3}(2x+x^2)^{3/2} + C$

1.7 $\arctan x^2 + C$

3.1 $\left(\frac{1}{2}\right)\left(\frac{1}{2}\right)x^2 + \frac{1}{2}\ln|x| + C$

3.2 $f(t) = 6\sqrt{t} + 3$

3.3 $2\arcsin x - \sqrt{1-x^2} + C$

3.4 10

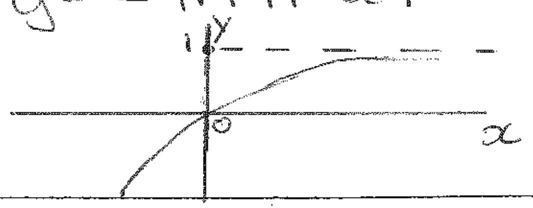
4. $\lim\limits_{x \to 0} y = e^2$

5. $\dfrac{\bar{a} \times \bar{b}}{|\bar{a} \times \bar{b}|} = \left\langle -\frac{1}{\sqrt{27}}, -\frac{1}{\sqrt{27}}, -\frac{5}{\sqrt{27}} \right\rangle$

6. See Proofs on ClickUP

7.1 $y = -\ln|1-x|$

7.2



7.3 Area =
$$-\int_{-1}^{0}(1-e^{-x})\,dx + \int_0^1 (1-e^{-x})\,dx$$

7.4
$$\frac{f(a)-1}{a-1} = e^{-a}$$
$$\Rightarrow \frac{-e^{-a}}{a-1} = e^{-a}$$
$$\Rightarrow a = 0 \Rightarrow f(a) = 0$$
$$\Rightarrow y = x$$

OR $f'(x) = e^{-x} \Rightarrow f'(a) = e^{-a}$

Eq of tangent through $(a, f(a))$ is:
$$y - f(a) = f'(a)(x-a)$$
$$\Rightarrow y - (1-e^{-a}) = e^{-a}(x-a)$$
which passes through $(1,1)$
$$\Rightarrow 1 - (1-e^{-a}) = e^{-a}(1-a)$$
$$\Rightarrow e^{-a} = e^{-a} - ae^{-a}$$
$$\Rightarrow -ae^{-a} = 0$$
$$\Rightarrow a = 0$$
$$\Rightarrow f(a) = 0$$
$\Rightarrow$ Eq of tangent is $y = x$